

| Natural language statement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Domain                   | GroundTruth                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
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| 8.19. Theorem. For a compact metric space $X$ , $H^s(X) = \sup \{H^s(C) : C \text{ subset } X \text{ is compact with } H^s(C) < \infty\}$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Geometric measure theory | <pre> theorem mattila_8_19_compact_subsets_of_finite_hausdorff_measure {X : Type u} [MetricSpace X] [CompactSpace X] [MeasurableSpace X] [BorelSpace X] {s : ℝ} (hs : 0 &lt; s) : muH[s] (Set.univ : Set X) =   C : Set X, ( _ : IsCompact C), ( _ : muH[s] C &lt; inf), muH[s] C := by sorry </pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 12.14. Theorem. Let $A$ be a Borel set in $\mathbb{R}^n$ . (1) If $\dim A > (n+1)/2$ , then $L_1(D(A)) > 0$ . (2) If $(n-1)/2 < \dim A < (n+1)/2$ , then $\dim D(A) > \dim A - (n-1)/2$ , where $D(A) = \{[x,y] : x,y \text{ in } A\}$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Geometric measure theory | <pre> theorem mattila_12_14_falconer_distance_set {n : ℕ} (hn : 2 &lt;= n) {A : Set (EuclideanSpace ℝ (Fin n))} (hA : MeasurableSet A) : let D := {r : ℝ   exists x in A, exists y in A, r = dist x y} (((n : ℝ&gt;=0inf) + 1) / 2 &lt; dimH A -&gt; 0 &lt; volume D) /\ (((n : ℝ&gt;=0inf) - 1) / 2 &lt; dimH A /\ dimH A &lt; ((n : ℝ&gt;=0inf) + 1) / 2 -&gt; dimH A - ((n : ℝ&gt;=0inf) - 1) / 2 &lt; dimH D) := by sorry </pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 6.1. Definition. The upper $s$ -dimensional density of $A$ at $x$ is $\Theta^s(A,x) = \limsup_{r \downarrow 0} (2r)^{-s} H^s(A \cap B(x,r))$ .<br>6.2. Theorem. Suppose $A$ subset $\mathbb{R}^n$ with $H^s(A) < \infty$ . (1) $2^s \Theta^s(A,x) \leq \Theta^s(A,x) \leq 1$ for $H^s$ almost all $x$ in $A$ . (2) If $A$ is $H^s$ measurable, $\Theta^s(A,x) = 0$ for $H^s$ almost all $x$ in $\mathbb{R}^n \setminus A$ .                                                                                                                                                                                                                                                                                                                                                        | Geometric measure theory | <pre> noncomputable def upperHausdorffDensity {n : ℕ} (s : ℝ) (A : Set (EuclideanSpace ℝ (Fin n))) (x : EuclideanSpace ℝ (Fin n)) : ℝ&gt;=0inf := limsup (fun r : ℝ =&gt; muH[s] (A inter closedBall x r) / ENNReal.ofReal ((2 ^ r) ^ s)) (nhds[&gt;] 0)  theorem mattila_6_2_hausdorff_density_estimates {n : ℕ} {s : ℝ} {A : Set (EuclideanSpace ℝ (Fin n))} (hs : 0 &lt; s) (hA : muH[s] A &lt; inf) : (forall^m x dmuH[s], x in A -&gt; ENNReal.ofReal (2 ^ (-s)) &lt;= upperHausdorffDensity s A x /\ upperHausdorffDensity s A x &lt;= 1) /\ (MeasurableSet A -&gt; forall^m x dmuH[s], x notin A -&gt; upperHausdorffDensity s A x = 0) := by sorry </pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 15.3. Definition. A set $E$ subset $\mathbb{R}^n$ is called $m$ -rectifiable if there exist Lipschitz maps $f_i : \mathbb{R}^m \rightarrow \mathbb{R}^n$ , $i = 1, 2, \dots$ , such that $H^m(E \setminus \bigcup_i f_i(\mathbb{R}^m)) = 0$ . A set $F$ subset $\mathbb{R}^n$ is called purely $m$ -unrectifiable if $H^m(E \text{ inter } F) = 0$ for every $m$ -rectifiable set $E$ . 18.1. Theorem. Let $A$ be an $H^m$ measurable subset of $\mathbb{R}^n$ with $H^m(A) < \infty$ . (1) $A$ is $m$ -rectifiable if and only if $H^m(P_V B) > 0$ for $\{n,m\}$ almost all $V$ in $G(n,m)$ whenever $B$ is an $H^m$ measurable subset of $A$ with $H^m(B) > 0$ . (2) $A$ is purely $m$ -unrectifiable if and only if $H^m(P_V A) = 0$ for $\{n,m\}$ almost all $V$ in $G(n,m)$ . | Geometric measure theory | <pre> def Grassmannian (n m : ℕ) := {V : Submodule ℝ (EuclideanSpace ℝ (Fin n)) // Module.finiteRank V = m}  def grassmannianAction {n m : ℕ} (Q : EuclideanSpace ℝ (Fin n) equivL[ℝ] EuclideanSpace ℝ (Fin n)) (V : Grassmannian n m) : Grassmannian n m := &lt;V.1.map Q.toLinearMap, (Q.finiteRank_map_eq V.1).trans V.2&gt;  def IsInvariantGrassmannianMeasure {n m : ℕ} [MeasurableSpace (Grassmannian n m)] ( : Measure (Grassmannian n m)) : Prop := IsProbabilityMeasure /\ forall Q : EuclideanSpace ℝ (Fin n) equivL[ℝ] EuclideanSpace ℝ (Fin n), (forall x,   Q x   =   x   -&gt; AEMeasurable (grassmannianAction (m := m) Q) /\ Measure.map (grassmannianAction (m := m) Q) =  def RectifiableSet (n m : ℕ) (E : Set (EuclideanSpace ℝ (Fin n))) : Prop := exists f : ℕ -&gt; EuclideanSpace ℝ (Fin m) -&gt; EuclideanSpace ℝ (Fin n), (forall j, exists K : NNReal, LipschitzWith K (f j)) /\ muH[(m : ℝ)] (E \ union j, range (f j)) = 0  def PurelyUnrectifiableSet (n m : ℕ) (A : Set (EuclideanSpace ℝ (Fin n))) : Prop := forall E : Set (EuclideanSpace ℝ (Fin n)), RectifiableSet n m E -&gt; muH[(m : ℝ)] (A inter E) = 0  theorem mattila_18_1_besicovitch_federer_projection {n m : ℕ} (hm : 0 &lt; m) (hmn : m &lt; n) ( : Measure (Grassmannian n m)) (h : IsInvariantGrassmannianMeasure ) {A : Set (EuclideanSpace ℝ (Fin n))} (hA : MeasurableSet A) (hAfin : muH[(m : ℝ)] A &lt; inf) : (RectifiableSet n m A &lt;-&gt; forall B : Set (EuclideanSpace ℝ (Fin n)), MeasurableSet B -&gt; B subset A -&gt; 0 &lt; muH[(m : ℝ)] B -&gt; forall^m V d, 0 &lt; muH[(m : ℝ)] ((fun x =&gt; V.1.orthogonalProjectionOnto x) '' B)) /\ (PurelyUnrectifiableSet n m A &lt;-&gt; </pre> |

| Natural language statement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Domain                   | GroundTruth                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                          | <pre> forall^m V d, muH[(m : R)] ((fun x =&gt; V.1.orthogonalProjectionOnto x) '' A) = 0) := by sorry </pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <p>7.7. Theorem. Let <math>A</math> subset <math>\mathbb{R}^n</math> and let <math>f: A \rightarrow \mathbb{R}^m</math> be a Lipschitz map. If <math>m &lt; s &lt; n</math>, then <math>\int_A H^s(s-m)(A \text{ inter } f^{-1}\{y\}) dL^m y \leq c(n,m) \text{Lip}(f)^m H^s(A)</math>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Geometric measure theory | <pre> theorem mattila_7_7_lipschitz_level_sets { n m : ℕ } :   exists c : ℝ ≥ 0, c &lt; inf /&amp; forall (s : ℝ) (A : Set (EuclideanSpace     ℝ (Fin n)))     (f : EuclideanSpace ℝ (Fin n) -&gt; EuclideanSpace ℝ (Fin m)) (K :       NNReal),     (m : ℝ) &lt; s /&amp; s &lt; n -&gt; LipschitzOnWith K f A -&gt;       upperIntegral volume (fun y =&gt; muH[s - m] (A inter f ^-1' {y})) &lt;=         c * (K : ℝ ≥ 0) ^ (m : ℝ) * muH[s] A := by sorry </pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <p>Definition. The <math>s</math>-dimensional Hausdorff content of <math>A</math> is <math>H^s_{\text{inf}}(A) = \inf \sum_i d(E_i)^s</math>, where the infimum is over all countable covers <math>A \subset \bigcup_i E_i</math>. 8.8. Theorem. Let <math>B</math> be a Borel set in <math>\mathbb{R}^n</math>. Then <math>H^s(B) &gt; 0</math> if and only if there exists <math>\mu</math> in <math>M(B)</math> such that <math>\mu(B(x,r)) &lt; r^s</math> for <math>x</math> in <math>\mathbb{R}^n</math> and <math>r &gt; 0</math>. Moreover, we can find <math>\mu</math> so that <math>\mu(B) &gt; c H^s_{\text{inf}}(B)</math>, where <math>c &gt; 0</math> depends only on <math>n</math>.</p>                                                                                                                  | Geometric measure theory | <pre> noncomputable def hausdorffContent {X : Type u} [PseudoMetricSpace X] (s : ℝ) (A : Set X) : ℝ ≥ 0 :=   U : ℕ -&gt; Set X, ( _ : A subset union i, U i),   sum' i, Metric.ediam (U i) ^ s  theorem mattila_8_8_frostman_lemma { n : ℕ } :   exists c : ℝ ≥ 0, 0 &lt; c /&amp; c &lt; inf /&amp;   forall (s : ℝ) (B : Set (EuclideanSpace ℝ (Fin n))),     0 &lt; s -&gt; MeasurableSet B -&gt;       (0 &lt; muH[s] B &lt;-&gt; exists mu : Measure (EuclideanSpace ℝ (Fin n)),         IsFiniteMeasure mu /&amp; IsFiniteMeasureOnCompacts mu /&amp;         Measure.InnerRegular mu /&amp;         mu != 0 /&amp; IsCompact mu.support /&amp; mu.support subset B /&amp;         forall x : EuclideanSpace ℝ (Fin n), forall r : ℝ, 0 &lt; r -&gt;           mu (closedBall x r) &lt; ENNReal.ofReal (r ^ s)) /&amp;       (0 &lt; muH[s] B -&gt; exists mu : Measure (EuclideanSpace ℝ (Fin n)),         IsFiniteMeasure mu /&amp; IsFiniteMeasureOnCompacts mu /&amp;         Measure.InnerRegular mu /&amp;         mu != 0 /&amp; IsCompact mu.support /&amp; mu.support subset B /&amp;         (forall x : EuclideanSpace ℝ (Fin n), forall r : ℝ, 0 &lt; r -&gt;           mu (closedBall x r) &lt; ENNReal.ofReal (r ^ s)) /&amp;         c * hausdorffContent s B &lt; mu B) := by sorry </pre>                                                          |
| <p>Definition. The Riesz <math>s</math>-energy of <math>\mu</math> is <math>I_s(\mu) = \int \int  x-y ^{-s} d\mu(x) d\mu(y)</math>. The Grassmannian <math>G(n,m)</math> is the space of <math>m</math>-dimensional linear subspaces of <math>\mathbb{R}^n</math>, and <math>\nu_{n,m}</math> is its orthogonally invariant probability measure. 9.7. Theorem. Let <math>\mu</math> be a Radon measure on <math>\mathbb{R}^n</math> with compact support and with <math>I_m(\mu) &lt; \infty</math>. Then <math>P_{\{V\}} \mu \ll H^m</math> for <math>\nu_{n,m}</math> almost all <math>V</math> in <math>G(n,m)</math> and <math>\int D(P_{\{V\}} \mu, u)^2 dH^m u</math> almost all <math>V</math> in <math>G(n,m)</math>, where <math>c</math> is a constant depending only on <math>n</math> and <math>m</math>.</p> | Geometric measure theory | <pre> noncomputable def rieszEnergy { n : ℕ } (s : ℝ) (mu : Measure (EuclideanSpace ℝ (Fin n))) : ℝ ≥ 0 :=   integral^~ x, (integral^~ y, (ENNReal.ofReal (dist x y)) ^ -1 ^ s dmu) dmu  def Grassmannian (n m : ℕ) :=   {V : Submodule ℝ (EuclideanSpace ℝ (Fin n)) // Module.finiteRank ℝ V = m}  def grassmannianAction { n m : ℕ } (Q : EuclideanSpace ℝ (Fin n) equivl[R] EuclideanSpace ℝ (Fin n)) (V : Grassmannian n m) : Grassmannian n m :=   &lt;V.1.map Q.toLinearMap, (Q.finiteRank_map_eq V.1).trans V.2&gt;  def IsInvariantGrassmannianMeasure { n m : ℕ } [MeasurableSpace (Grassmannian n m)] ( _ : Measure (Grassmannian n m)) : Prop :=   IsProbabilityMeasure /&amp;   forall Q : EuclideanSpace ℝ (Fin n) equivl[R] EuclideanSpace ℝ (Fin n),     (forall x,   Q x   =   x  ) -&gt; AEMeasurable (grassmannianAction (m := m) Q) /&amp;     Measure.map (grassmannianAction (m := m) Q) =  theorem mattila_9_7_projection_energy { n m : ℕ } ( _ : Measure (Grassmannian n m)) (h : IsInvariantGrassmannianMeasure) :   exists c : ℝ ≥ 0, c &lt; inf /&amp;   forall mu : Measure (EuclideanSpace ℝ (Fin n)),     IsFiniteMeasureOnCompacts mu -&gt; Measure.InnerRegular mu -&gt; IsCompact mu.support -&gt;   rieszEnergy (m : ℝ) mu &lt; inf -&gt;   (forall^m V d,     Measure.map (fun x =&gt; V.1.orthogonalProjectionOnto x) mu &lt;= </pre> |

## Natural language statement

## Domain

## GroundTruth

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| <p>Definition. The Grassmannian <math>G(n,k)</math> is the space of <math>k</math>-dimensional linear subspaces of <math>R^n</math>, and <math>_{\{n,k\}}</math> is its orthogonally invariant probability measure. 10.10. Theorem. Let <math>m &lt; t &lt; n</math> and let <math>A</math> subset <math>R^n</math> be a Borel set with <math>0 &lt; H^t(A) &lt; \infty</math>. Then for all <math>W</math> in <math>G(n,n-m)</math>, <math>H^t(t-m)(A \cap W) &lt; \infty</math> for <math>H^m</math> almost all <math>a</math> in <math>W^{\text{bot}}</math>, and for <math>_{\{n,n-m\}}</math> almost all <math>W</math> in <math>G(n,n-m)</math>, <math>H^m(\{a \text{ in } W^{\text{bot}} : \dim(A \cap W) = t-m\}) &gt; 0</math>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Geometric measure theory | <pre> def Grassmannian (n m : N) := {V : Submodule R (EuclideanSpace R (Fin n)) // Module.finiteRank R V = m}  def grassmannianAction {n m : N} (Q : EuclideanSpace R (Fin n) equivl[R] EuclideanSpace R (Fin n)) (V : Grassmannian n m) : Grassmannian n m := &lt;V.1.map Q.toLinearMap, (Q.finiteRank_map_eq V.1).trans V.2&gt;  def IsInvariantGrassmannianMeasure {n m : N} [MeasurableSpace (Grassmannian n m)] ( : Measure (Grassmannian n m)) : Prop := IsProbabilityMeasure /\ forall Q : EuclideanSpace R (Fin n) equivl[R] EuclideanSpace R (Fin n), (forall x,   Q x   =   x  ) -&gt; AEMeasurable (grassmannianAction (m := m) Q) /\ Measure.map (grassmannianAction (m := m) Q) =  theorem mattila_10_10_plane_sections {n m : N} {t : R} {A : Set (EuclideanSpace R (Fin n))} ( : Measure (Grassmannian n (n - m))) (h : IsInvariantGrassmannianMeasure ) (hmt : (m : R) &lt; t /\ t &lt; n) (hA : MeasurableSet A) (hAfinite : muH[t] A &lt; inf) (hApos : 0 &lt; muH[t] A) : let slice := fun (W : Grassmannian n (n - m)) (a : W.1) =&gt; A inter {x   x - (a : EuclideanSpace R (Fin n)) in W.1} (forall W : Grassmannian n (n - m), forall^m a d(muH[(m : R)] : Measure W.1), (muH[t - m] : Measure (EuclideanSpace R (Fin n))) (slice W a) &lt; inf) /\ forall^m W : Grassmannian n (n - m) d, 0 &lt; muH[(m : R)] {a : W.1   dimH (slice W a) = ENNReal.ofReal (t - m)} := by sorry </pre> |
| <p>14.10. Theorem. Let <math>s</math> be a positive number. Suppose that there exists a Radon measure <math>\mu</math> on <math>R^n</math> such that the density <math>\Theta_s(\mu, a)</math> exists and is positive and finite in a set of positive <math>\mu</math> measure. Then <math>s</math> is an integer.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Geometric measure theory | <pre> theorem mattila_14_10_marstrand_density_integer {n : N} {s : R} {mu : Measure (EuclideanSpace R (Fin n))} (hs : 0 &lt; s) (hmu : IsFiniteMeasureOnCompacts mu /\ Measure.InnerRegular mu) (hdensity : exists E : Set (EuclideanSpace R (Fin n)), 0 &lt; mu E /\ forall x in E, exists theta : R&gt;=0inf, 0 &lt; theta /\ theta &lt; inf /\ Tendsto (fun r : R =&gt; mu (closedBall x r) / ENNReal.ofReal ((2 * r) ^ s)) (nhds[&gt;] 0) (nhds theta)) : exists m : N, s = m := by sorry </pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <p>15.3. Definition. A set <math>E</math> subset <math>R^n</math> is called <math>m</math>-rectifiable if there exist Lipschitz maps <math>f_i : R^m \rightarrow R^n</math>, <math>i = 1, 2, \dots</math>, such that <math>H^m(E \setminus \bigcup_i f_i(R^m)) = 0</math>. 15.7. Definition. We say that a subset <math>E</math> of <math>R^n</math> is <math>m</math>-linearly approximable if for <math>H^m</math> almost all <math>a</math> in <math>E</math> the following holds: if <math>\epsilon</math> is a positive number, there are positive numbers <math>r_0</math> and <math>\lambda</math> and an affine <math>m</math>-plane <math>W</math> such that <math>a</math> in <math>W</math> and for any <math>0 &lt; r &lt; r_0</math>, <math>H^m(E \cap B(a, r)) \geq \lambda r^m</math> for <math>x</math> in <math>W</math> inter <math>B(a, r)</math>, and <math>H^m(E \cap B(a, r) \setminus W(\epsilon r)) &lt; \epsilon r^m</math>. 15.17. Definition. Let <math>A</math> subset <math>R^n</math>, <math>a</math> in <math>R^n</math> and <math>V</math> in <math>G(n,m)</math>. We say that <math>V</math> is an approximate tangent <math>m</math>-plane for <math>A</math> at <math>a</math> if <math>\Theta_s^m(A, a) &gt; 0</math> and for all <math>0 &lt; s &lt; 1</math>, <math>r^s H^m(A \cap B(a, r) \setminus X(a, V, s)) \rightarrow 0</math> as <math>r \rightarrow 0</math>. 15.19. Theorem. Let <math>E</math> be an <math>H^m</math> measurable subset of <math>R^n</math> with <math>H^m(E) &lt; \infty</math>. Then the following are equivalent: (1) <math>E</math> is <math>m</math>-rectifiable. (2) <math>E</math> is <math>m</math>-linearly approximable. (3)</p> | Geometric measure theory | <pre> def Grassmannian (n m : N) := {V : Submodule R (EuclideanSpace R (Fin n)) // Module.finiteRank R V = m}  def RectifiableSet (n m : N) (E : Set (EuclideanSpace R (Fin n))) : Prop := exists f : N -&gt; EuclideanSpace R (Fin m) -&gt; EuclideanSpace R (Fin n), (forall j, exists K : NNReal, LipschitzWith K (f j)) /\ muH[(m : R)] (E \ union j, range (f j)) = 0  def LinearlyApproximableSet (n m : N) (E : Set (EuclideanSpace R (Fin n))) : Prop := forall^m a dmuH[(m : R)].restrict E, forall eta : R, 0 &lt; eta -&gt; exists V : Grassmannian n m, exists r0 c : R, 0 &lt; r0 /\ 0 &lt; c /\ forall r : R, 0 &lt; r -&gt; r &lt; r0 -&gt; (forall x, x - a in V.1 -&gt; x in closedBall a r -&gt; ENNReal.ofReal (c * r ^ m) &lt;= muH[(m : R)] (E inter closedBall x </pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

Natural language statement

Domain

GroundTruth

For  $H^m$  almost all  $a$  in  $E$  there is a unique approximate tangent  $m$ -plane for  $E$  at  $a$ . (4) For  $H^m$  almost all  $a$  in  $E$  there is some approximate tangent  $m$ -plane for  $E$  at  $a$ .

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(eta * r))) /\
muH[(m : R)] (E inter closedBall a r inter
  {x | eta * r <= infDist (x - a) (V.1 : Set (EuclideanSpace R (Fin
n)))))) <
  ENNReal.ofReal (eta * r ^ m)

def IsApproximateTangentPlane {n m : N} (E : Set (EuclideanSpace R (Fin n)))
  (a : EuclideanSpace R (Fin n)) (V : Grassmannian n m) : Prop :=
  0 < upperHausdorffDensity (m : R) E a /\
  forall s : R, 0 < s -> s < 1 ->
    Tendsto (fun r : R =>
      muH[(m : R)] (E inter closedBall a r inter
        {x | s * dist x a <= infDist (x - a)
          (V.1 : Set (EuclideanSpace R (Fin n))})) /
        ENNReal.ofReal (r ^ (m : R))) (nhds[>] 0) (nhds 0)

theorem mattila_15_19_rectifiability_tangent_planes
  {n m : N} {E : Set (EuclideanSpace R (Fin n))}
  (hE meas : MeasurableSet E) (hE finite : muH[(m : R)] E < inf) :
  List.TFAE [RectifiableSet n m E, LinearlyApproximableSet n m E,
    forall^m a dmuH[(m : R)].restrict E,
      exists! V : Grassmannian n m, IsApproximateTangentPlane (m := m) E
        a V,
    forall^m a dmuH[(m : R)].restrict E,
      exists V : Grassmannian n m, IsApproximateTangentPlane (m := m) E a
      V] := by
  sorry
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